Bayesian Lasso

Lasso, regularization as Bayesian priors

Bayesian versus Frequentist Inference

	Bayesian	Frequentist
Probability	subjective degree of belief	limiting frequency
Goal	analyze beliefs	create procedures with frequency guarantees
θ	random variable	fixed
X	random variable	random variable
Use Bayes' theorem?	Yes. To update beliefs.	Yes, if it leads to procedure with good frequentist behavior. Otherwise no.

Table 1: Bayesian versus Frequentist Inference

Posterior belief using Bayes' rule

- Prior and likelihood

$$\theta \sim \pi$$
$$\{X_1, \ldots, X_n\} | \theta \sim P_\theta.$$

- Posterior

$$\pi(\theta | X_1, \ldots, X_n) = \frac{\mathcal{L}(\theta; X_1, \ldots, X_n)\pi(\theta)}{\int_\theta \mathcal{L}(\theta; X_1, \ldots, X_n)\pi(\theta) d\theta} \propto \mathcal{L}(\theta)\pi(\theta)$$

Credible interval

- The frequentist treats the likelihood as just a function of θ.
- The Bayesian (weights and) normalizes the likelihood and interprets it as a distribution over the support of θ.
- Once we have a posterior distribution, we can construct what are called credible sets; they are the Bayesian analogue of confidence sets but are quite different.

$$\int_{C_\alpha} \pi(\theta|X_1,\ldots,X_n)d\theta = 1-\alpha.$$

Bernstein-von Mises theorem

- In fixed-dimensional problems, under the assumption that the prior is continuous, and (strictly) positive in a neighborhood around the true θ, the posterior is close to a Gaussian (mean at MLE),

$$\left\| \pi(\theta|X_1,\ldots,X_n) - N\left(\widehat{\theta}_n, \frac{I(\widehat{\theta}_n)^{-1}}{n}\right) \right\|_{\mathrm{TV}} \to 0,$$

- The distance between the two distributions is the total-variation distance,

$$\|p-q\|_{\mathrm{TV}} = \frac{1}{2}\int |p(x)-q(x)|dx.$$

Where do priors come from?

- In fixed dimension, large sample-size problems, under some conditions Bayesian procedures will behave like standard frequentist inference.
- Some might also choose priors based on the data; this is known as empirical Bayes. This is often a good idea but some would consider it to not strictly adhere to the Bayesian philosophy.
- Some have argued for what are called non-informative priors, i.e. priors that somehow capture complete ignorance about the parameter.

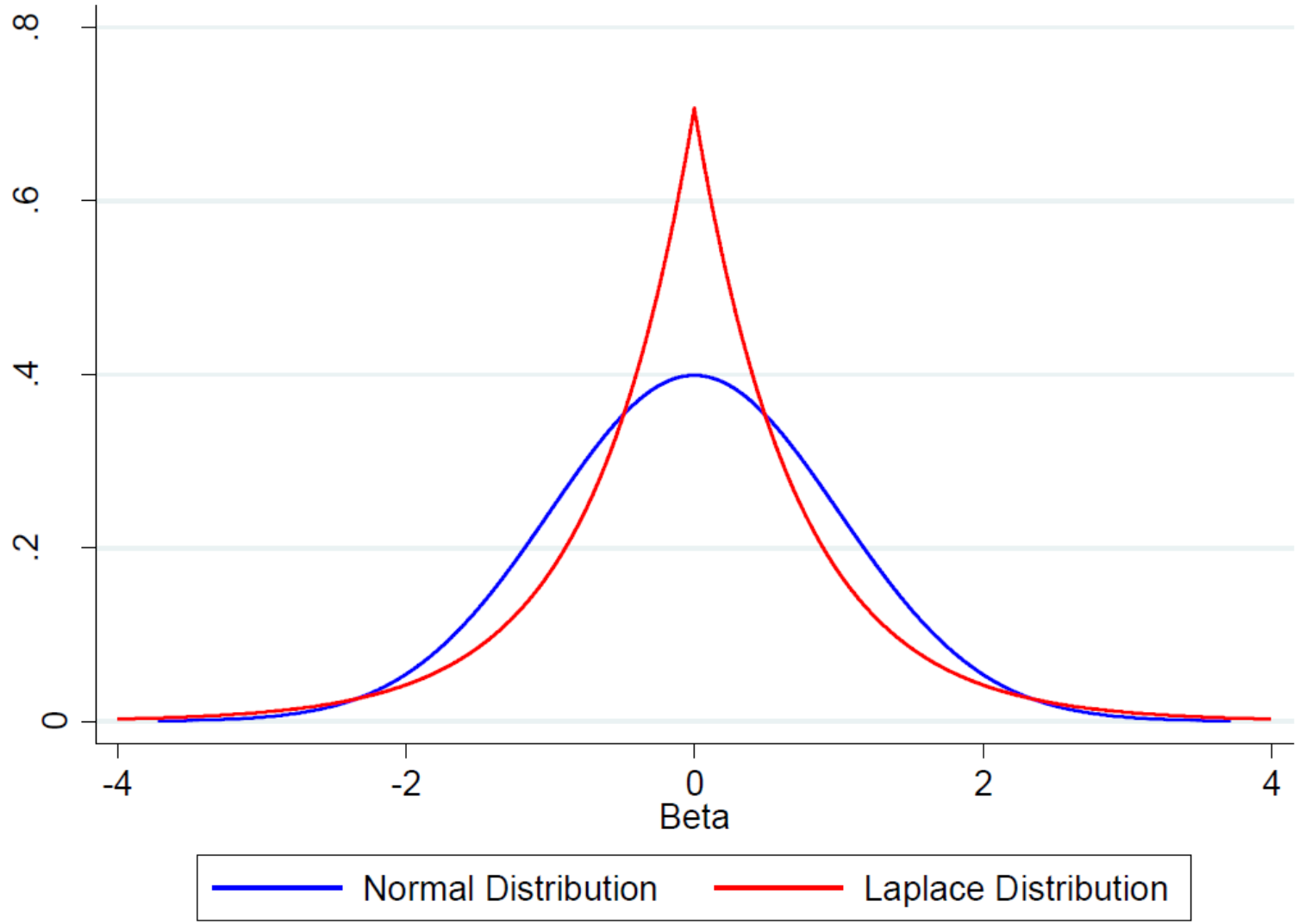
Priors = Regularizers?

- The LASSO regression estimator or Ridge regression estimator are just the posterior mode with either a Laplace prior or a Gaussian prior (respectively).
- The argument is that many sensible frequentists procedures (ones with strong guarantees) are just posterior summaries with particular priors.

Regularizer

- We can fit a model containing all p predictors using a technique that constrains or regularizes the coefficient estimates, or equivalently, that shrinks them towards zero.
- Least Absolute Selection and Shrinkage Operator (Lasso)

Standard Normal distribution and Laplace distribution



Ridge regression

- The least squares procedure estimates parameters by minimizing

$$\text{RSS} = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

- The ridge regression coefficient estimates minimize where $\lambda \geq 0$ is a tuning parameter

$$\sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

$$\underset{\beta}{\text{minimize}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 \leq s,$$

Lasso

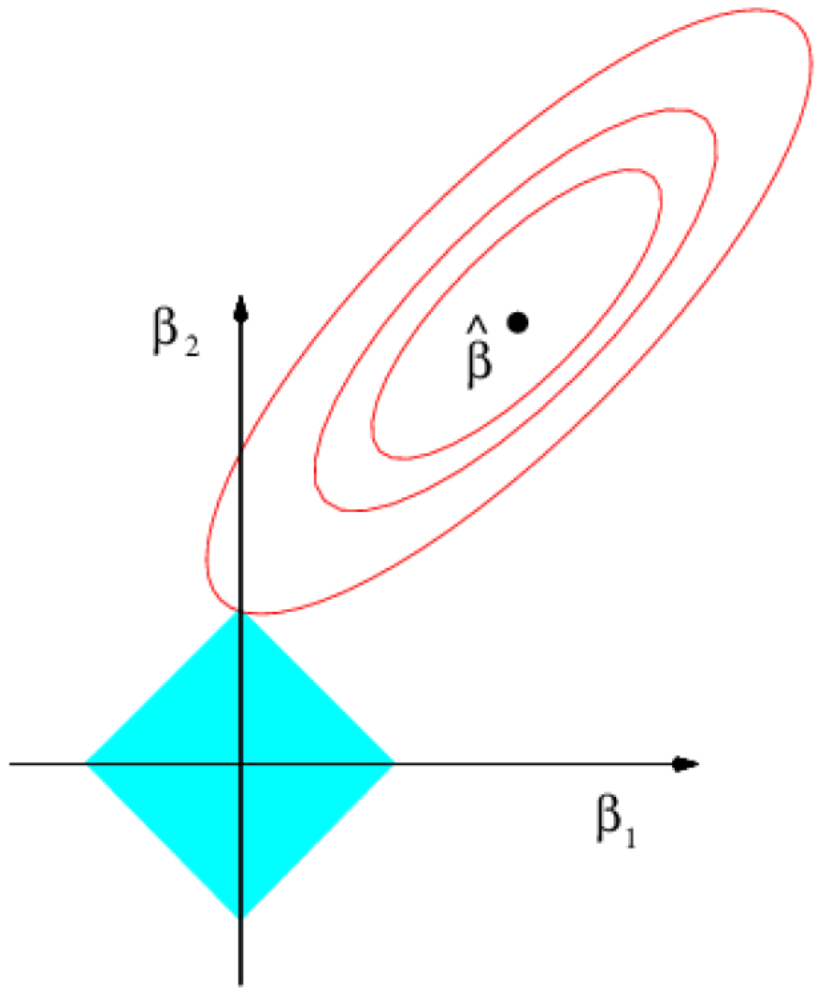
- The lasso coefficients minimize

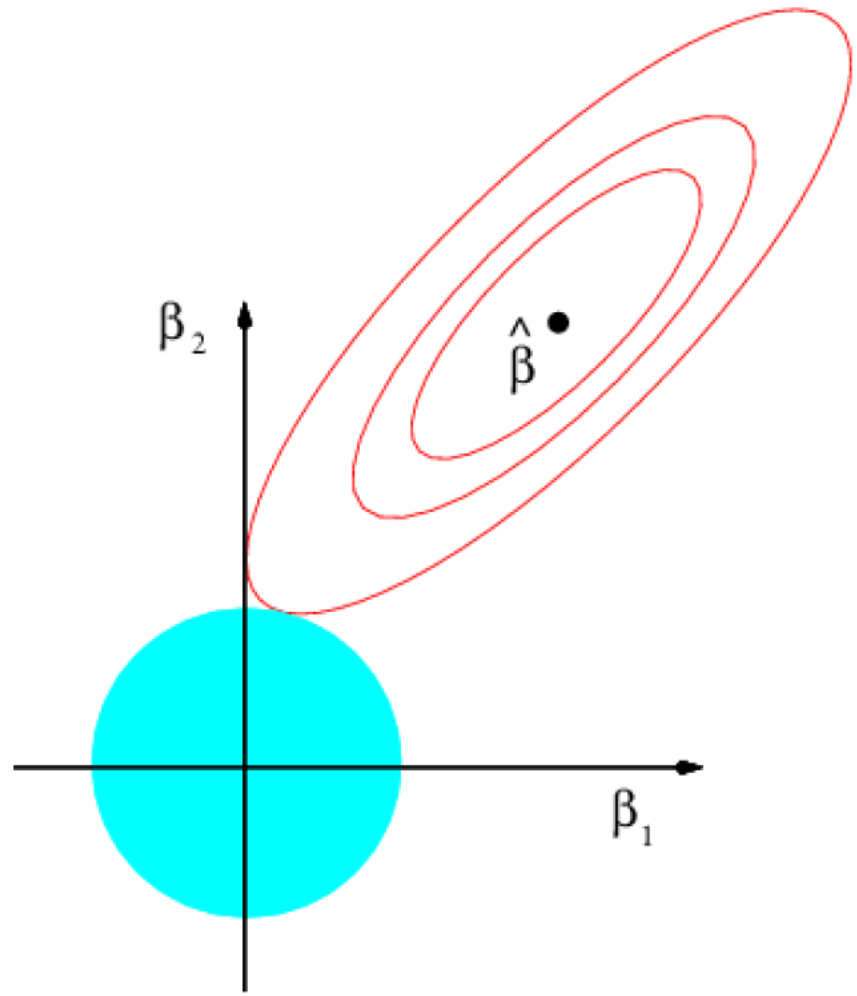
$$\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j x_{ij}\right)^2+\lambda\sum_{j=1}^{p}|\beta_j|$$

- The L1 penalty has the effect of forcing some of the coefficient estimates to be exactly equal to zero when the tuning parameter is sufficiently large.

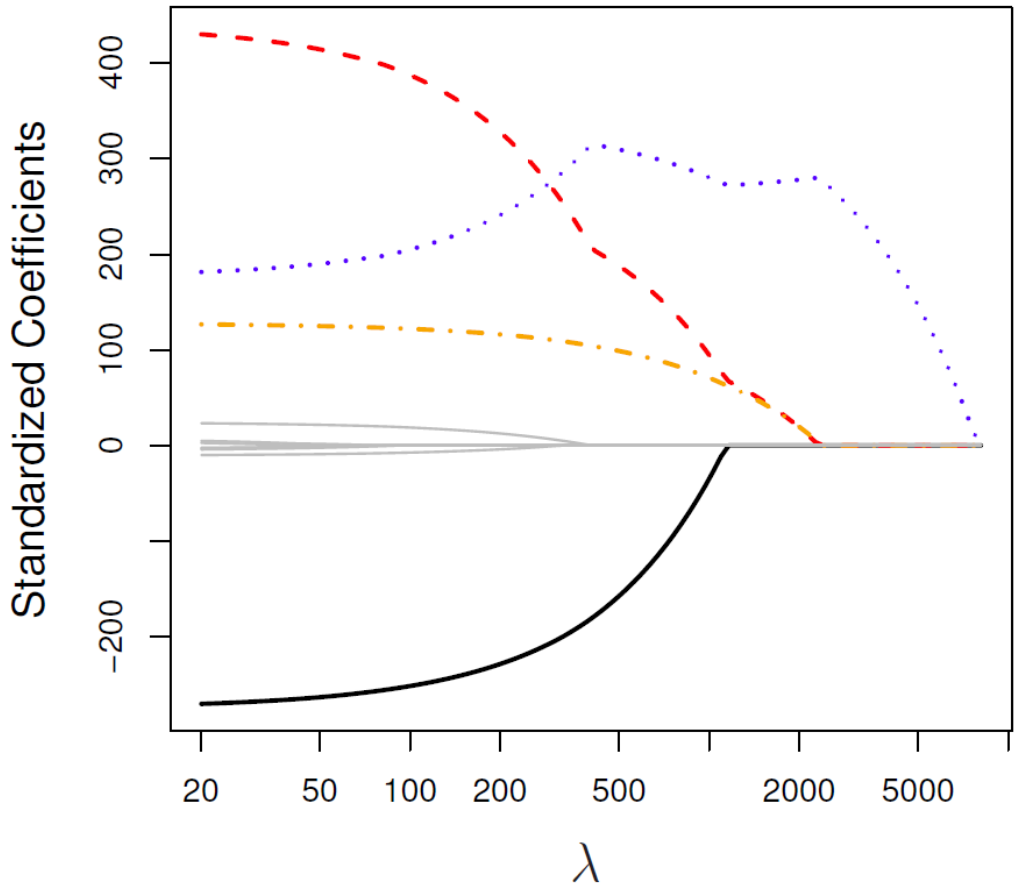
$$\underset{\beta}{\text{minimize}}\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j x_{ij}\right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p}|\beta_j|\le s$$

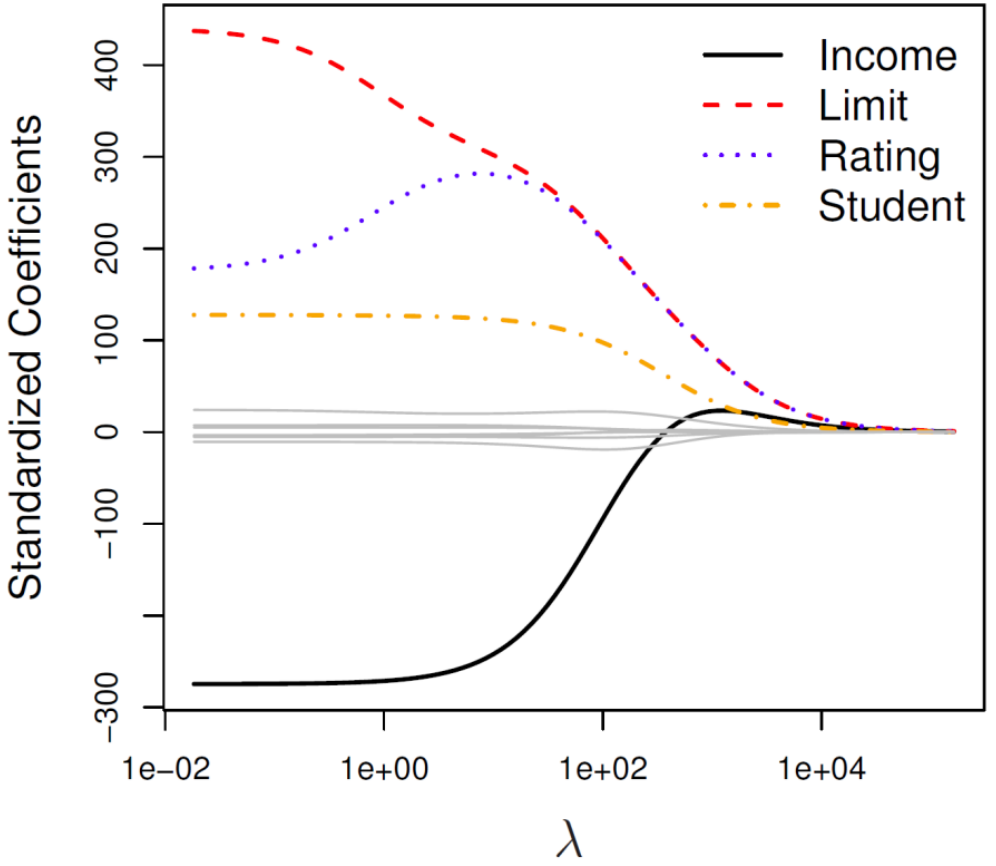
L_1 vs L_2 penalty





Lasso vs Ridge





L_1-norm regularization

- Minimize

$$J(w) = \sum_{i=1}^{n} \ell(y_i, w^\top x_i) + \lambda \|w\|_1$$

Error on data + Regularization

$$\hat{\beta}^{\text{lasso}} = \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \|y - X\beta\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

$$= \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \underbrace{\|y - X\beta|_2^2}_{\text{Loss}} + \lambda \underbrace{\|\beta\|_1}_{\text{Penalty}}$$

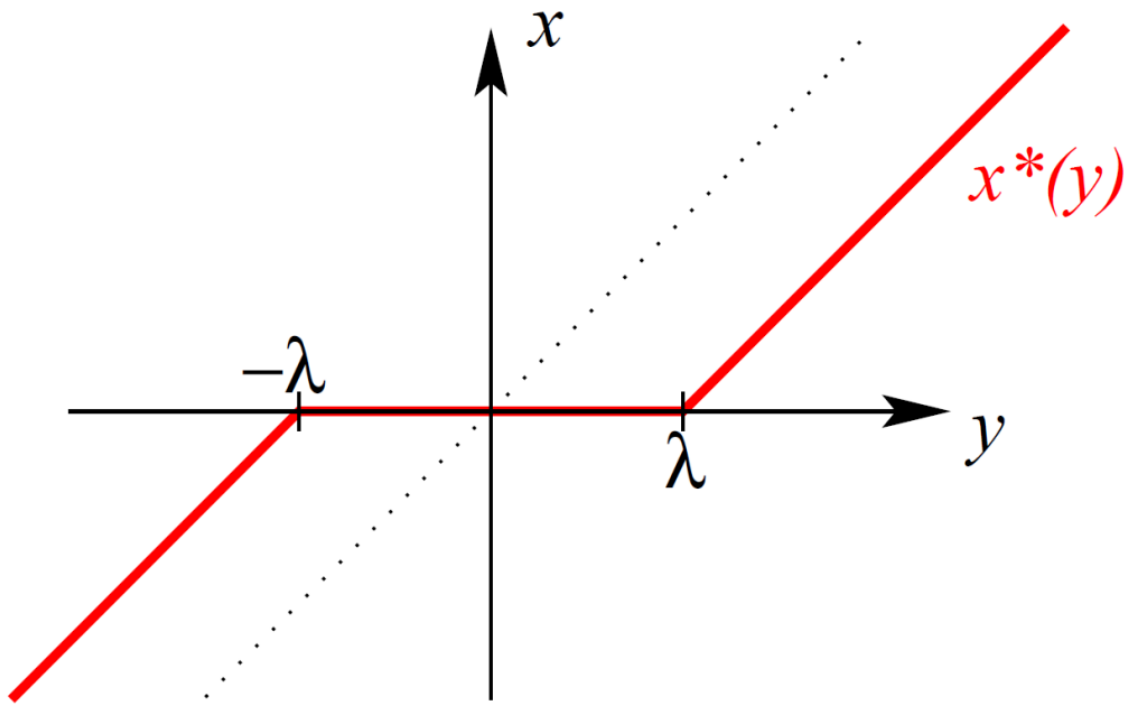
Why L1-norm leads to sparsity?

- Consider a quadratic problem

$$\min_{x \in \mathbb{R}} \frac{1}{2} x^2 - xy + \lambda |x|$$

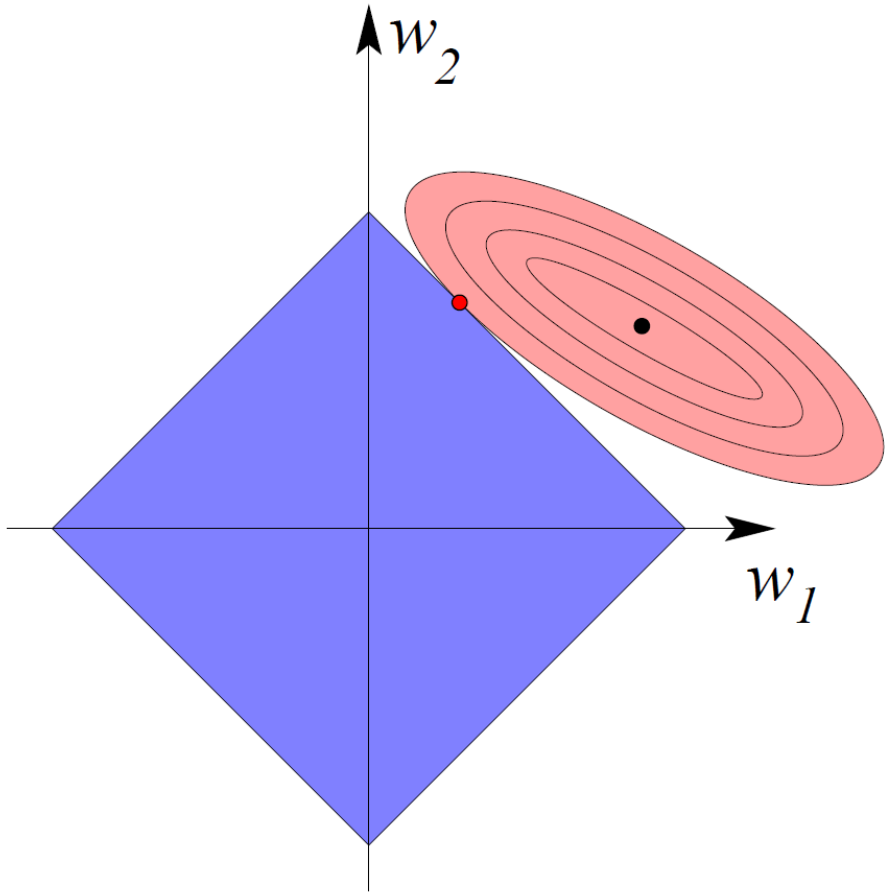
- The solution is soft thresholding

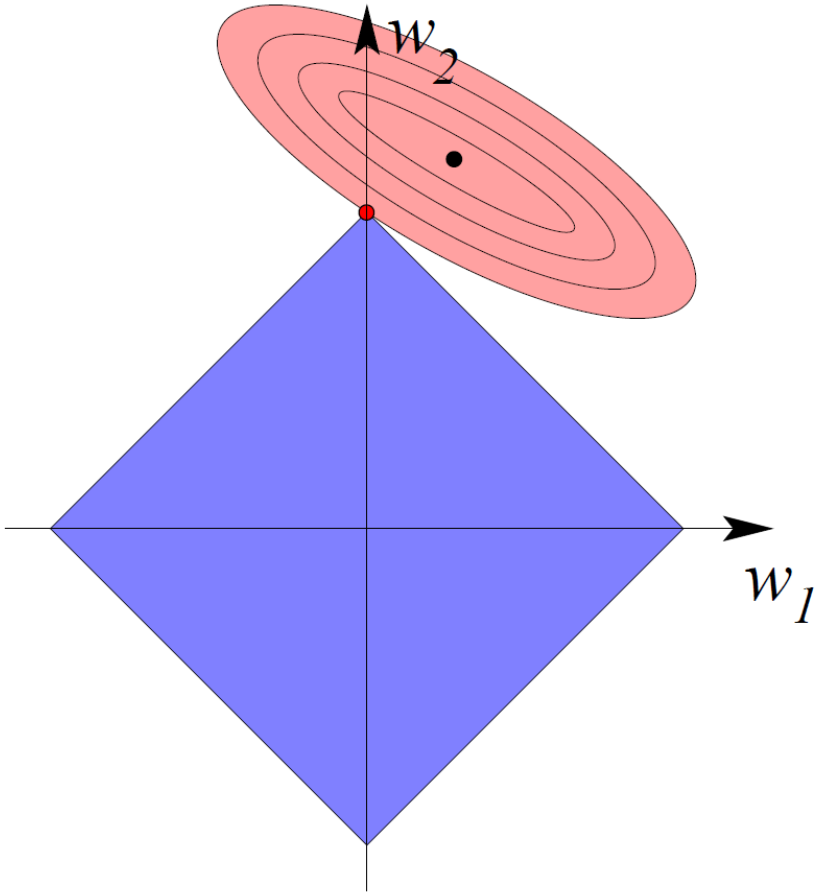
$$x^* = \text{sign}(y)(|y| - \lambda)_+$$



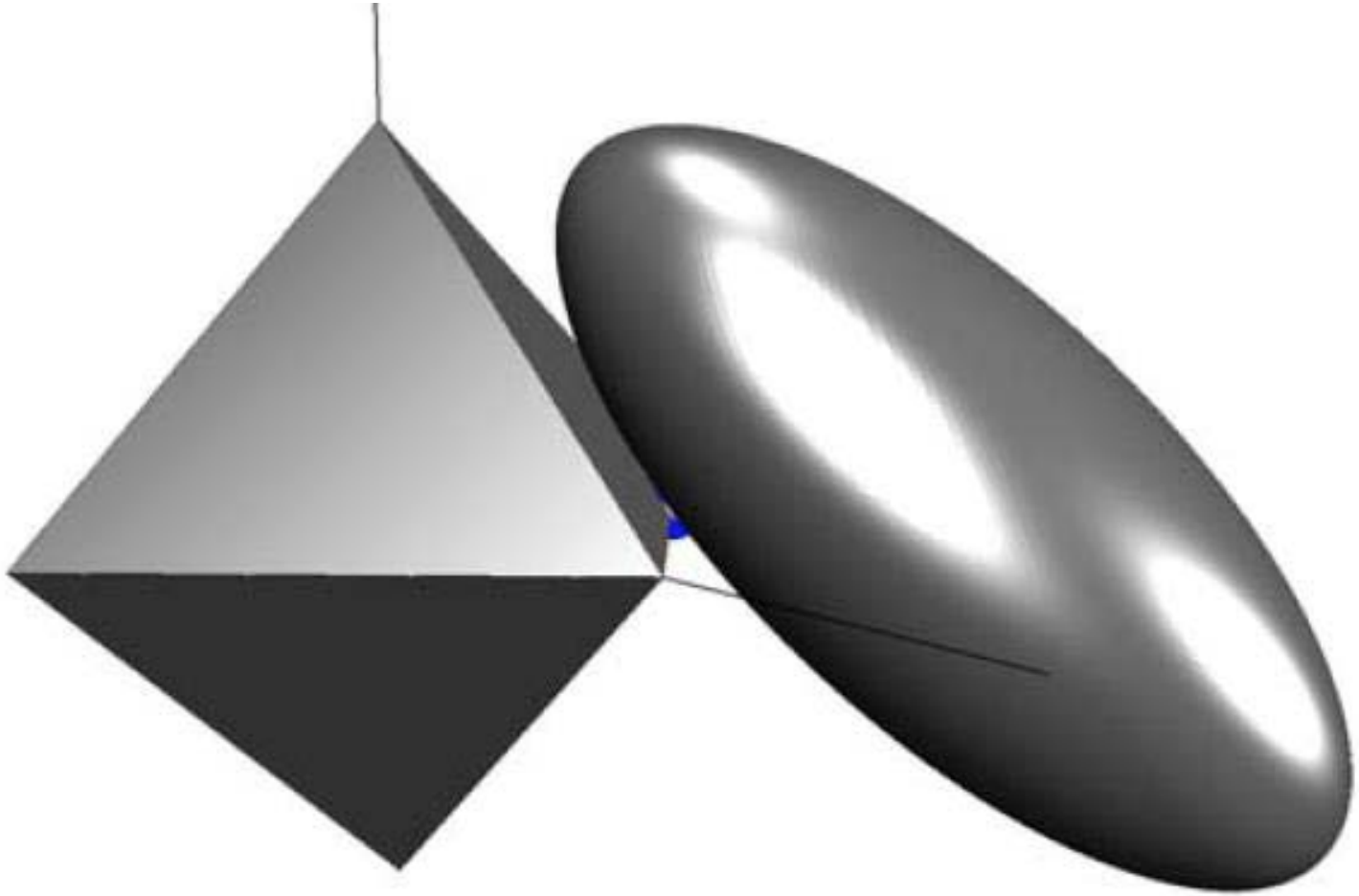
Penalizing is “equivalent” to constraining

- Change the tuning parameter to make the diamond smaller or bigger

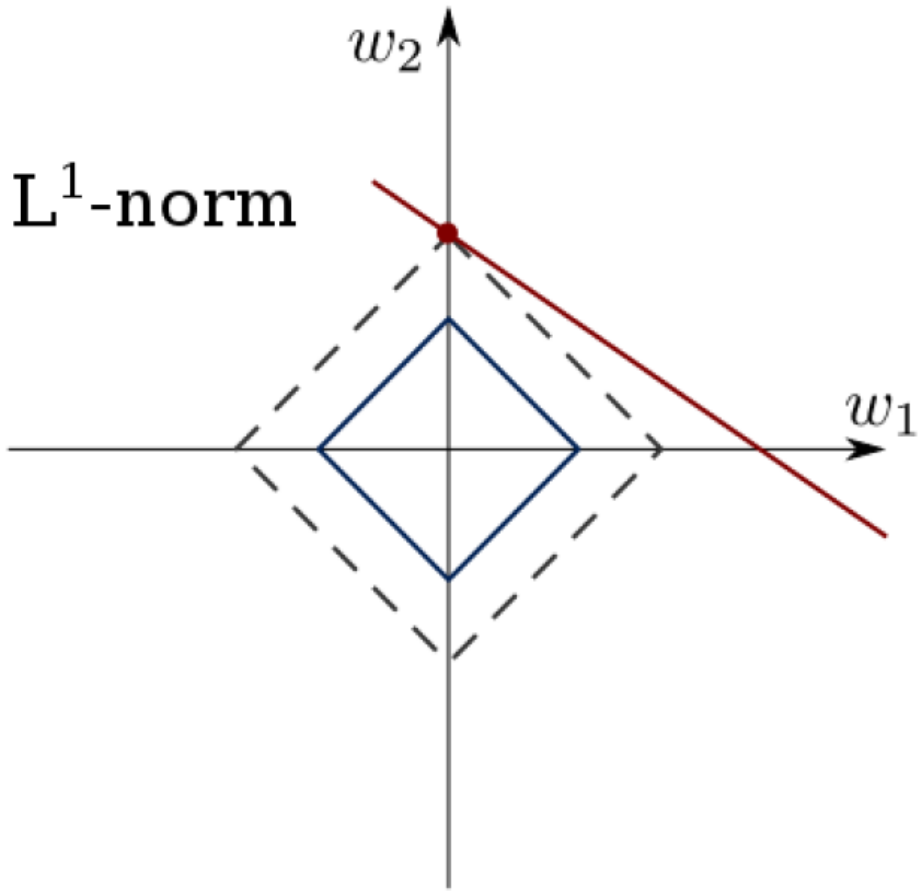


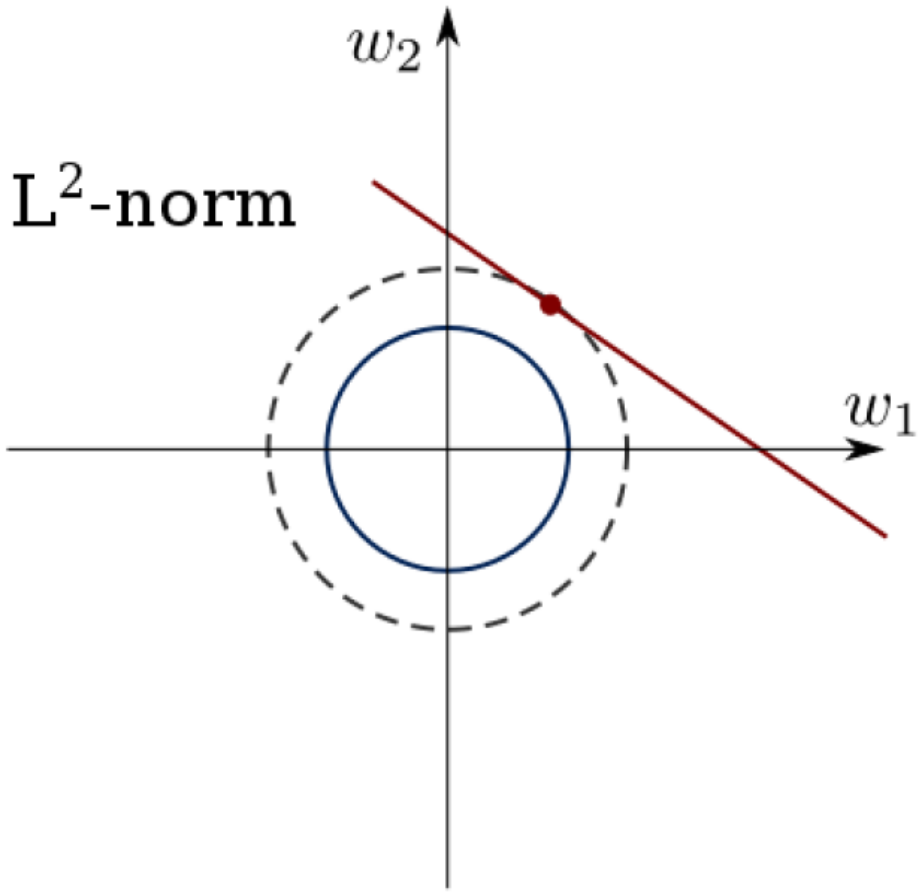


Lasso 3D view



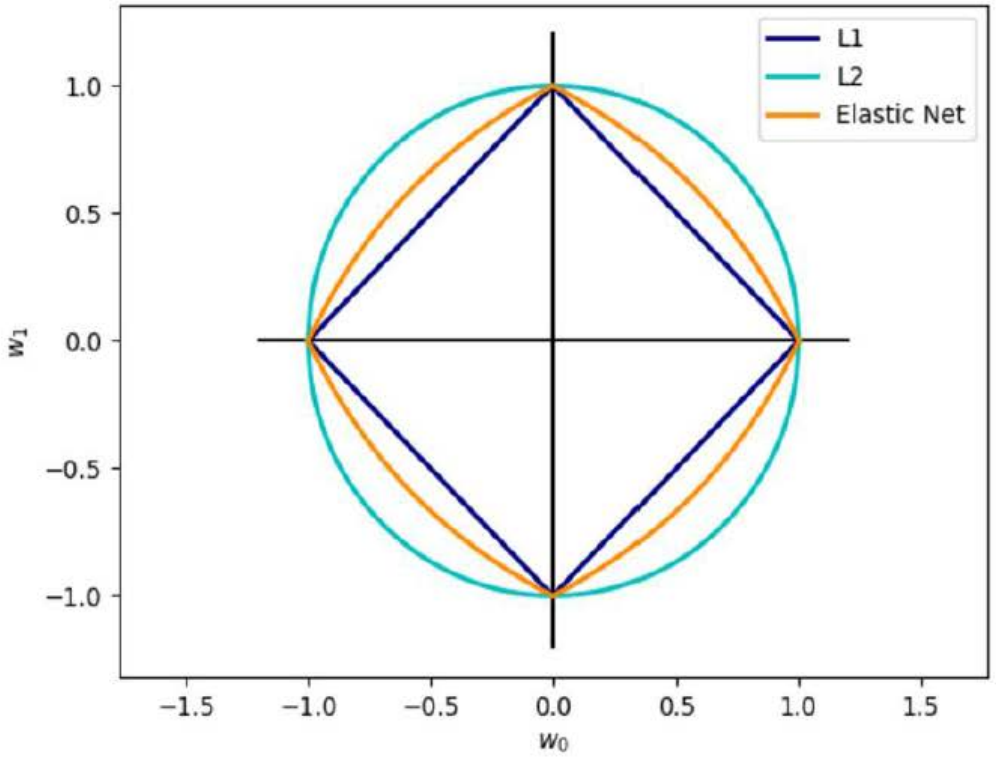
Lasso and ridge regression

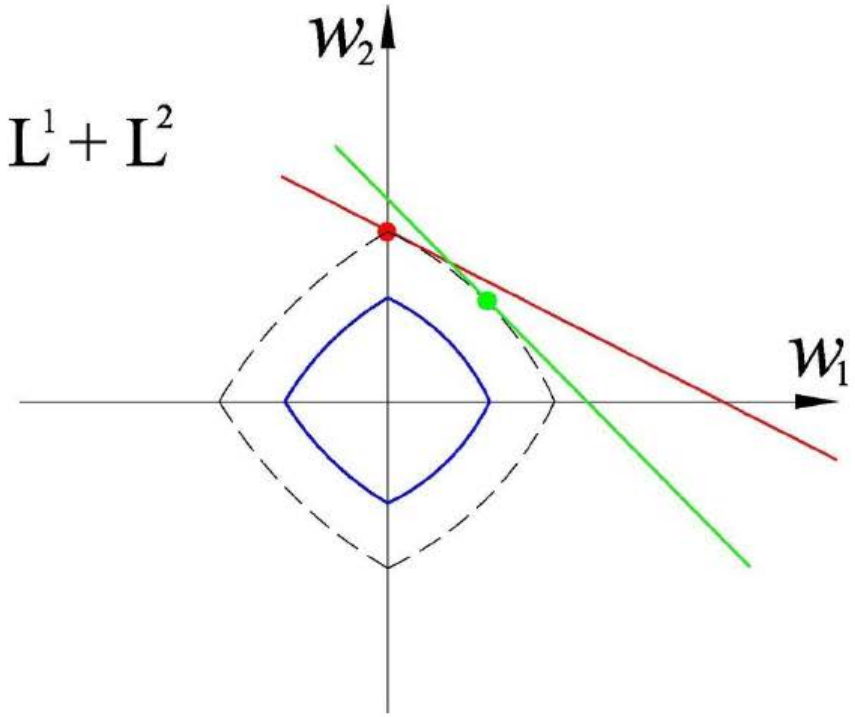




Elastic net

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}}(\|y - X\beta\|^2 + \lambda_2 \|\beta\|^2 + \lambda_1 \|\beta\|_1).$$

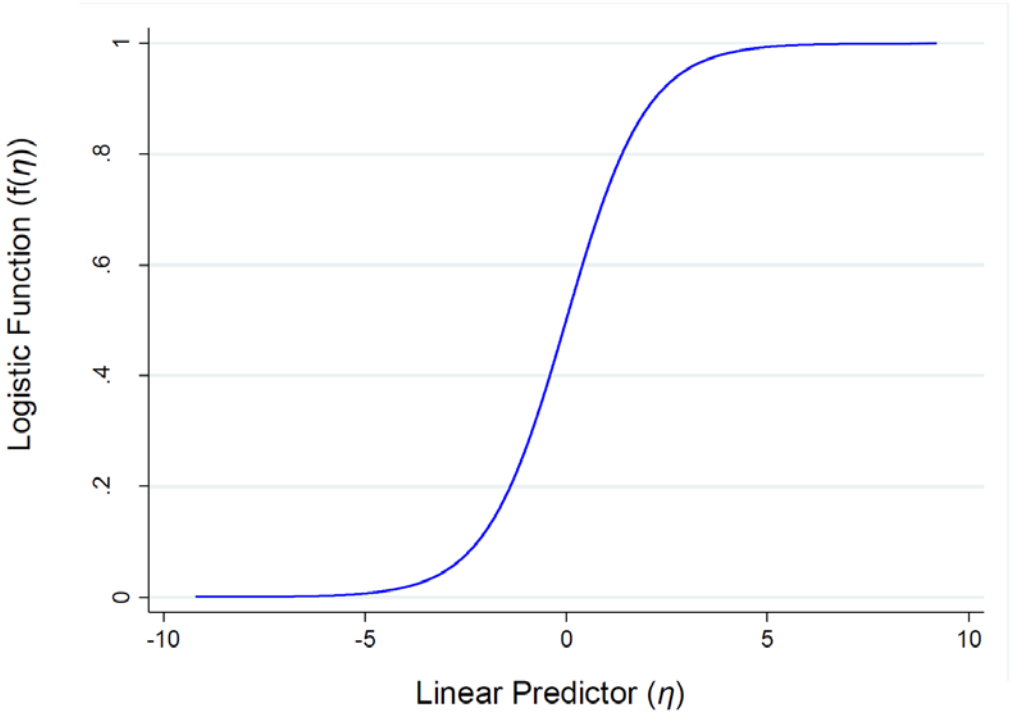




LASSO in Logistic Model

- Logistic model

$$\pi_i = P(Y_i = 1|\boldsymbol{x}_i, \boldsymbol{\beta}) = f(\eta_i|\boldsymbol{x}_i, \boldsymbol{\beta}) = \frac{\exp(\eta_i)}{1+\exp(\eta_i)} = \frac{1}{1+\exp(-\eta_i)}$$



Lasso Logistic Regression Model

- Success probability

$$\pi_i = P(Y_i = 1|\boldsymbol{x}_i', \boldsymbol{\beta}) = \frac{1}{1 + \exp(-\boldsymbol{\beta}^T \boldsymbol{x}_i')}$$

- Likelihood

$$p(y_1, \dots, y_n|\boldsymbol{\beta}) = \prod_{i=1}^{n} (\pi_i)^{y_i} (1 - \pi_i)^{1-y_i}$$

$$= \prod_{i=1}^{n} \left(\frac{1}{1 + \exp(-\boldsymbol{\beta}^T \boldsymbol{x}_i')} \right)^{y_i} \left(\frac{\exp(-\boldsymbol{\beta}^T \boldsymbol{x}_i')}{1 + \exp(-\boldsymbol{\beta}^T \boldsymbol{x}_i')} \right)^{1-y_i}$$

Log likelihood

$$l(\boldsymbol{\beta}) = \ln\left(L(\boldsymbol{\beta}|y_1, \dots, y_n)\right)$$

$$= \sum_{i=1}^{n}\left[y_i \ln\left(\frac{1}{1+\exp\left(-\boldsymbol{\beta}^T \boldsymbol{x}_i'\right)}\right) + (1-y_i)\ln\left(\frac{\exp(-\boldsymbol{\beta}^T \boldsymbol{x}_i')}{1+\exp\left(-\boldsymbol{\beta}^T \boldsymbol{x}_i'\right)}\right)\right]$$

$$= -\sum_{i=1}^{n}[(1-y_i)\boldsymbol{\beta}^T \boldsymbol{x}_i' + \ln(1+\exp(-\boldsymbol{\beta}^T \boldsymbol{x}_i'))]$$

Constrained maximization equation

$$l(\boldsymbol{\beta})_c = -\sum_{i=1}^{n}[(1 - y_i)\boldsymbol{\beta}^T\boldsymbol{x}_i' + \ln(1 + \exp(-\boldsymbol{\beta}^T\boldsymbol{x}_i'))] - \lambda\sum_{k=1}^{p}|\beta_k|$$

- Laplace (double exponential) distribution with mean 0, where the probability density function with mean μ and variance $2/\lambda^2$ is

$$f(\beta_k) = \frac{\lambda}{2}\exp(-\lambda|\beta_k - \mu|)$$

$$\ln\left(\prod_{k=1}^{p}\frac{\lambda}{2}\exp(-\lambda|\beta_k|)\right) = p\ln(\lambda) - p\ln(2) - \lambda\sum_{k=1}^{p}|\beta_k|$$

$$\propto -\lambda\sum_{k=1}^{p}|\beta_k|$$

Bayesian hierarchical model

- DE (double exponential) distribution represents a Laplace distribution with mean 0 and variance parameter λ

$$y_i | \beta_0, \beta_1, \ldots, \beta_p, \lambda \sim Ber(\pi_i)$$

$$logit(\pi_i) = \beta_0 + \sum_{k=1}^{p} \beta_k x_{ik}$$

$$\beta_k | \lambda \sim DE(\beta_k | 0, \lambda)$$

Posterior

- The posterior probability density function of the model parameters

$$p(\boldsymbol{\beta}|y_1, \dots, y_n) \propto p(y_1, \dots, y_n|\boldsymbol{\beta})p(\boldsymbol{\beta})$$

$$= \left[\prod_{i=1}^{n} \left(\frac{1}{1+\exp(-\boldsymbol{\beta}^T \boldsymbol{x}_i')}\right)^{y_i} \left(\frac{\exp(-\boldsymbol{\beta}^T \boldsymbol{x}_i')}{1+\exp(-\boldsymbol{\beta}^T \boldsymbol{x}_i')}\right)^{1-y_i}\right] \cdot \left[\prod_{k=1}^{p} \frac{\lambda}{2} \exp(-\lambda|\beta_k|)\right]$$

$$\Rightarrow \ln\left(p(\boldsymbol{\beta}|y_1, \dots, y_n)\right) \propto -\sum_{i=1}^{n} [(1 - y_i)\boldsymbol{\beta}^T \boldsymbol{x}_i' + \ln(1+\exp(-\boldsymbol{\beta}^T \boldsymbol{x}_i'))] - \lambda \sum_{k=1}^{p} |\beta_k|$$